

PROJECT SENTINEL | DAY 1 ARCHITECTURE & BUILD REPORT

Virtual Intrusion Detection & Prevention System (VIDS/IPS)

1. Executive Summary & Core Working Mechanics

Project Sentinel was designed and built from scratch today as a fully operational, cloud-integrated Network Intrusion Detection & Prevention System (NIDS/IPS). The system functions as an automated security perimeter guard, continuously inspecting incoming and outgoing packet headers in real time and identifying reconnaissance patterns without interrupting regular traffic.

- **Packet Interception (Scapy):** Sits at the network boundary inspecting TCP/UDP headers (Source IP, Destination IP, Target Ports, and TCP flags).
- **Sliding Time-Window Engine:** Tracks rapid knocking patterns. If any unique IP targets ≥ 50 ports in ≤ 3 seconds, it mathematically flags a Port Scan Reconnaissance attack.
- **Multi-Threat Logic:** Expanded to identify DoS activity by tracking TCP SYN flag bursts (≥ 40 SYNs in ≤ 2 seconds).
- **Active IPS Mitigation:** Elevated from passive IDS to active IPS by executing OS-level firewall blocking commands (netsh on Windows, iptables on Linux) to drop attacking traffic.
- **Cloud Persistence (Supabase):** Dispatched telemetry logs directly to managed PostgreSQL using REST APIs and secured with Row Level Security (RLS).
- **Live Operations Console (Streamlit):** Delivered real-time telemetry streaming, metric counters, threat breakdown bar graphs, and CSV audit exporting.

2. Pipeline Architecture & Tech Stack

Subsystem	Technology	Engineering Function
Network Boundary Sniffer	Python / Scapy 2.x	Raw packet capture, frame decoding, loopback and Ethernet sniffing.
Threat Logic Engine	In-Memory Heuristics	Sliding window state-tracking algorithm for port and flag frequency.
Cloud Incident Store	Supabase (PostgreSQL)	Remote persistence, RLS security policies, and ISO8601 timestamps.
Security Operations UI	Streamlit & Pandas	Real-time reactive dashboard, severity filtering, and forensic CSV export.
Automated Mitigation	Windows Netsh / Firewall	Active prevention module generating OS packet drop rules.
Verification Harness	Python Sockets (Test)	Custom attack simulator verifying heuristic threshold triggers.

3. Industry-Standard Resume Documentation

- **Impact Summary:** Engineered a cloud-backed Network Intrusion Detection and Prevention System (NIDS/IPS) in Python leveraging Scapy for real-time raw packet telemetry and automated incident response.
- **Detection Algorithm:** Designed sliding time-window state machines to catch automated reconnaissance scans (50 ports in 3s) and TCP SYN floods, cutting anomalous traffic detection time to under 100ms.
- **Cloud & Visualization:** Integrated managed PostgreSQL storage via Supabase REST API and built an interactive SOC incident dashboard using Streamlit for live telemetry analytics and CSV auditing.